

Education

Bachelor of Technology – Software Engineering

CGPA: 7.3

2022 – 2026 (Expected)

Experience

AI/ML Engineer Intern

Jul 2025 – Aug 2025

- Developed a **multimodal skin lesion classification system** combining dermoscopic images and patient metadata for early cancer detection.
 - Designed a **CNN + MLP fusion architecture** using Keras Functional API, extracting image features with EfficientNetB0 and processing tabular metadata using dense neural networks.
 - Performed extensive preprocessing including **data balancing, feature scaling, label encoding, and image augmentation**.
 - Achieved **97–98% classification accuracy** on a multimodal skin lesion dataset.
 - Evaluated performance using **confusion matrix, precision, recall, F1-score, and ROC-AUC metrics**.
 - Improved model robustness by fusing clinical metadata such as age, gender, and lesion site with visual features.
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Projects

Explainable Multi-Modal AI for Cardiovascular Disease Prediction (Ongoing)

- Building a multimodal deep learning pipeline integrating **EHR, MRI, ECG, genomic, and clinical text data** using HL7 FHIR standards.
 - Implemented modality-specific encoders including **LSTM for EHR, CNN/ResNet for MRI, and MLP for genomic data** with late-fusion strategies.
 - Applied **Explainable AI techniques** such as SHAP, Captum Integrated Gradients, and Grad-CAM to improve transparency in model decisions.
 - Integrated **LLM-based systems** to generate clinician-friendly diagnostic summaries.
 - Used distributed computing frameworks for scalable multimodal preprocessing and training.
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Multimodal Skin Lesion Classification System

- Built a binary classifier distinguishing **benign vs malignant skin lesions** using combined image and tabular data.
 - Developed models using **EfficientNetB0, TensorFlow/Keras, scikit-learn, and Pandas**.
 - Implemented fusion strategies to improve diagnostic accuracy.
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Banana Quality Classification System

- Developed an AI-based image classification system to determine banana quality.

- Used **CNN models with OpenCV preprocessing and Python ML libraries**.
 - Built a simple web interface using **Streamlit** for interactive predictions.
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Technical Skills

Programming:

Python, C, C++

Machine Learning / AI:

TensorFlow, Keras, PyTorch, Scikit-learn, CNNs, LSTM, Multimodal Learning, Explainable AI (SHAP, Grad-CAM, Captum)

Web / UI:

HTML, CSS, Streamlit, Django

Data & Visualization:

Pandas, NumPy, Matplotlib, Seaborn, Tableau

Databases:

MySQL

Tools & Platforms:

Linux (Ubuntu), Virtual Machines, VS Code, Git

Productivity Tools:

MS Excel

Achievements & Activities

- Achieved **91.8% in Class XII and 95.6% in Class X**.
- Ranked in the **top 2 percentile in a national engineering entrance examination** and qualified for the advanced stage.
- **Course Topper** in an online Environmental Science course.
- Completed certifications in **Python programming and Internet of Things** with high scores.
- Mentored multiple students through a **government mentorship initiative**.
- Solved **500+ programming problems** across competitive programming platforms.